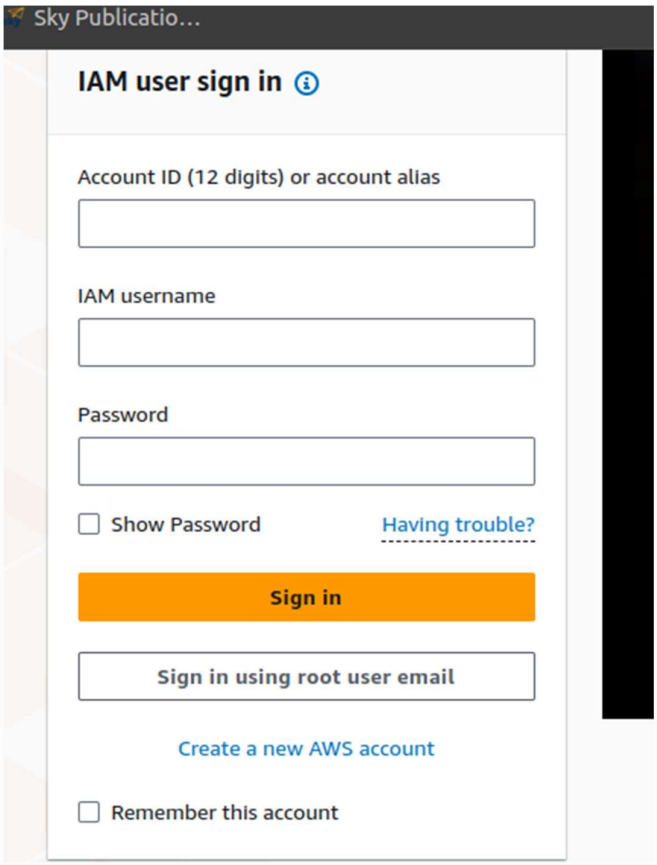
Cloud Computing

Exercise - 2 Total points: 10

Introduction to Infrastructure as a Service

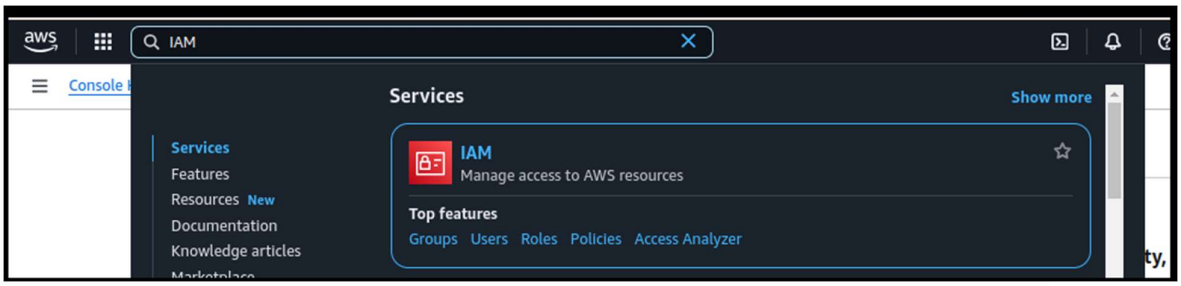
In this practice session you learn how to access cloud services that we will be using in the rest of the course. We would be using AWS Public Cloud. For this you need to create an AWS Account.

- Use the url: console.aws.amazon.com
  Scroll down and you would find the form as shown below, where you will find “create a new AWS account”
- While creating the account it would ask for your credit or debit card for user verification purposes and would deduct $1 from your account. Also, it would ask for an identity proof for validating the user again. (We gave PAN, probably you can give student id, voter id etc.)
- Make sure you create an account that is a free account and don’t opt for Developer or any other category which has subscription charges. With a free account (FREE TIER) you will be provided with limited amount of usage for certain services, which will be helpful in completion of our tasks.

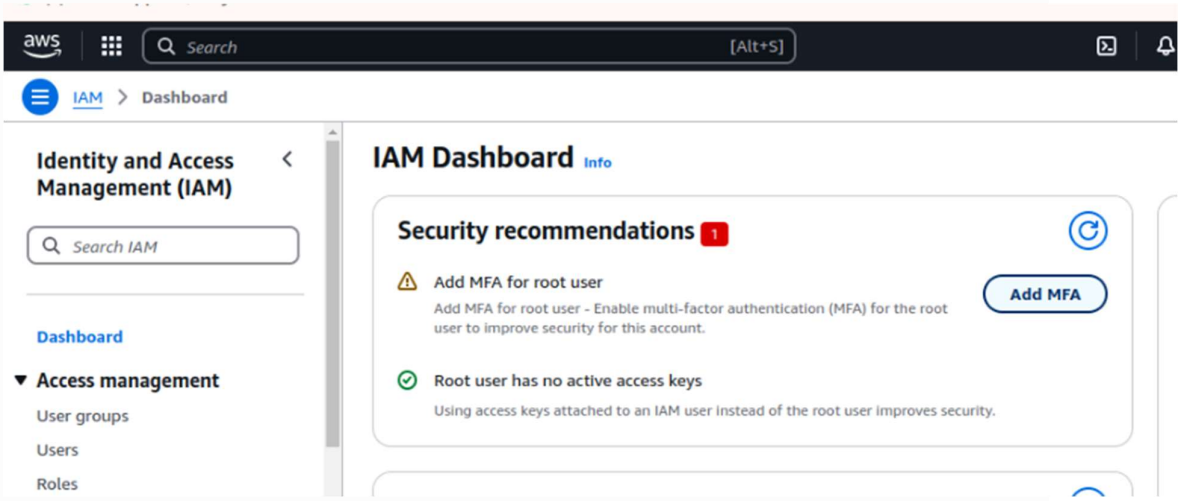


Verify that an AWS account is successfully created and get yourself familiarized with the AWS Console or UI. You will work with these things in the coming tasks.

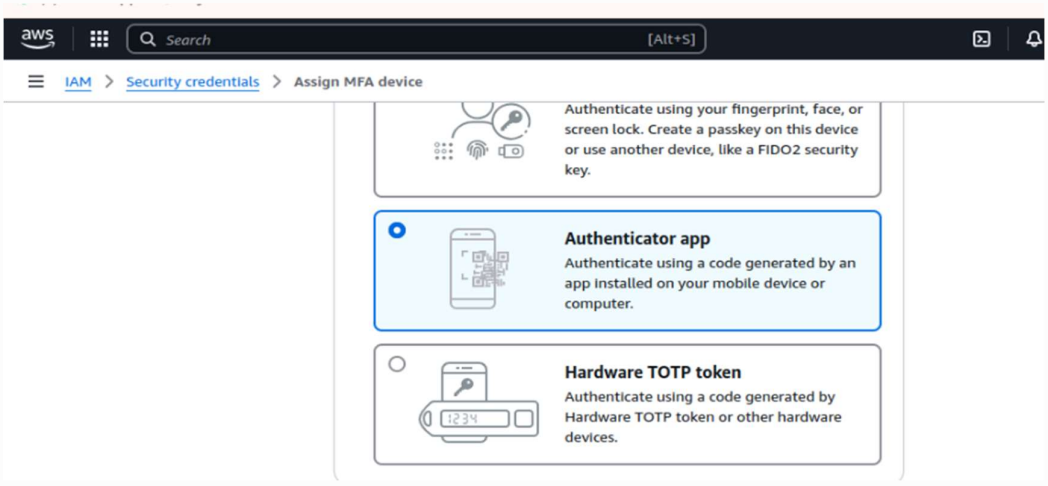
1. Go to IAM Dashboard and set up MFA (Multi Factor Authentication) to your root account. It would need you to install an application like Authenticator on your mobile for secure logging into your account.



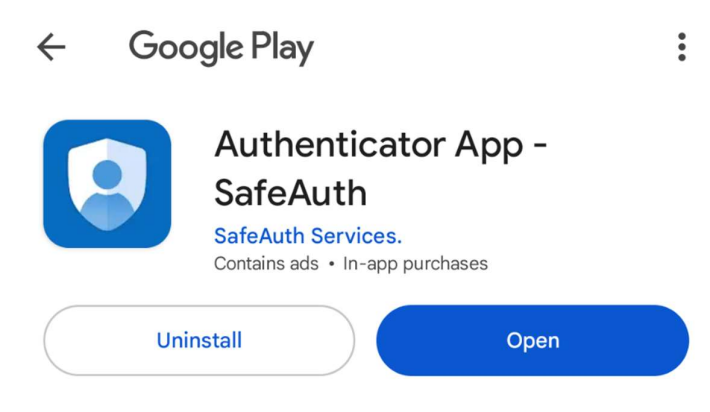
Note: You can search for IAM service in the search bar on the top left.



IAM-> Security->Assign MFA Device.

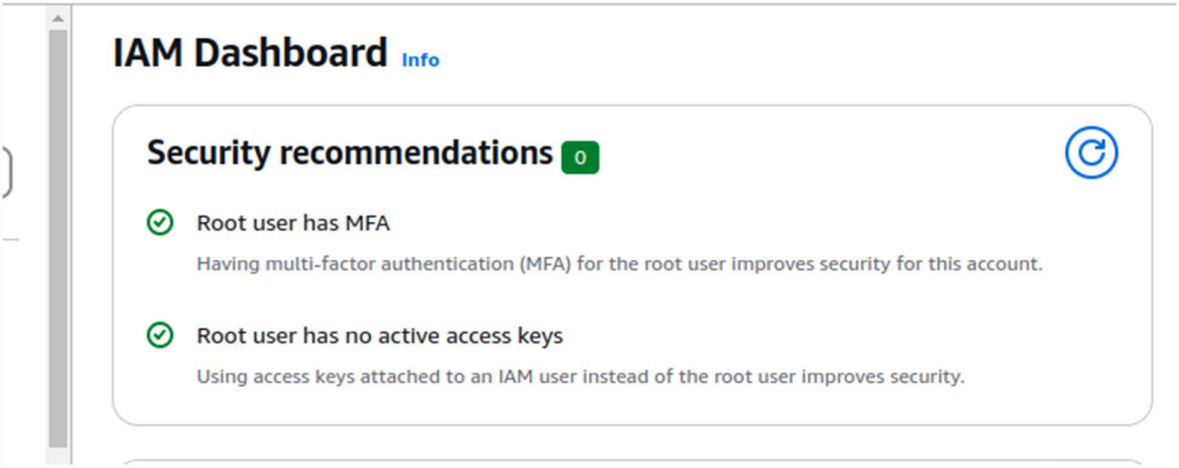


For Authenticator App you could install the following app on your



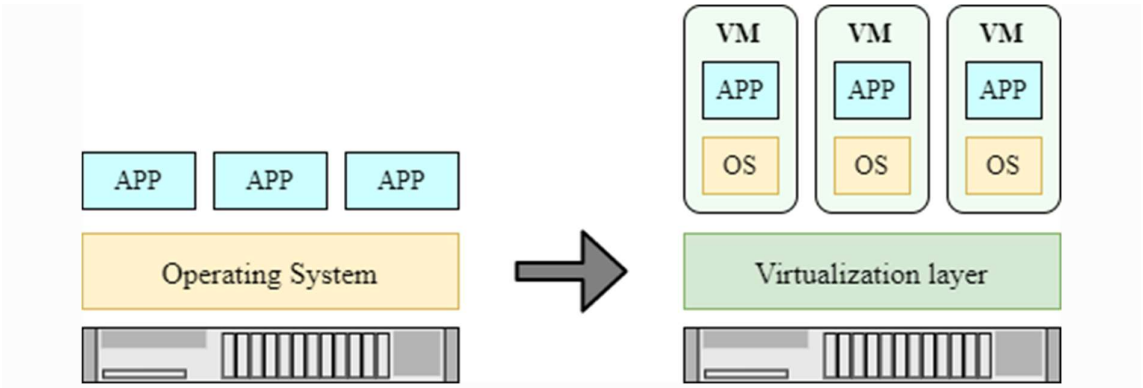
Once you open the app, you can click on “Add Project”, it will try to scan the QR Code which would be displayed on the AWS IAM Console, as part of your Authenticator App’s next step. It would then ask you to enter the passcode generated for two times. Then your MFA would be setup. Later every time when you sign in to your AWS Console, it would ask to enter the Passcode to login to the console.

If the MFA is set up successfully you could see the screen like below



Introduction

Infrastructure as a Service (IaaS) is a type of Cloud computing service, in which Virtualized computing resources are provided to users over the internet. In comparison to using physical servers, computing resources can be provisioned on-demand and in real-time and applications running on the same hardware can be separated into different secure environments, each containing their own OS, software libraries and kernels.



Working with the IaaS model of Cloud usually consists of the following steps:

1. Register an account to access the cloud services
2. Select appropriate virtual machine image to run (Ubuntu, Debian, Windows, etc.)
3. Start a new instance of the selected virtual machine image. Login into the instance as a root user over the internet and configure it to meet your requirements. I.e. install needed software, upload your own application, perform any required configuration actions as you would do in any real computer.
4. As you will lose all your work when instance will be terminated -- you have three options on how to persist the changes you made:
    1. Save all your configuration steps to a script that will launch and configure the instance automatically for you.
    2. Bundle a new image from your running instance and next time launch your custom image.
    3. Save the running instance as a snapshot, and next time launch new instances from there.

First option is more flexible as you can easily change the script than bundle a new image if something changes. Second and third options are simpler to use once you have stable configuration or when launching a large number of instances.

In this exercise we are working on the AWS cloud platform.

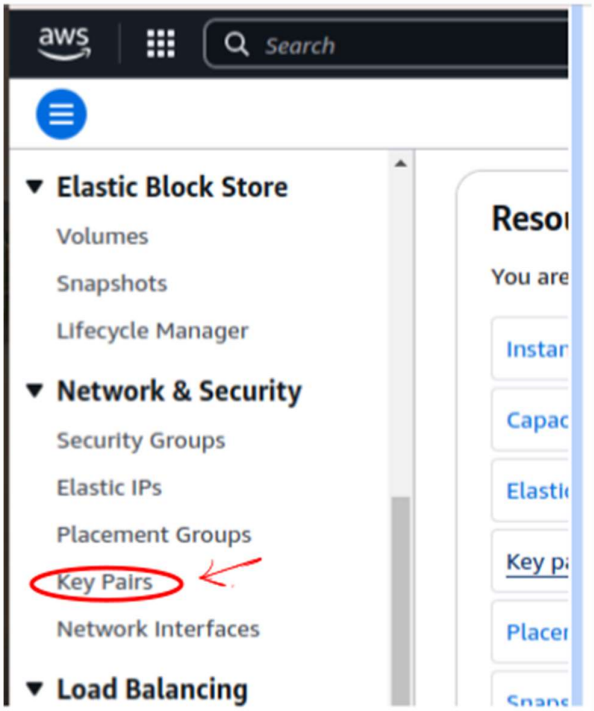
Exercise 2.1. Accessing the cloud services

In this exercise you will log into AWS Cloud and create a secure access key.

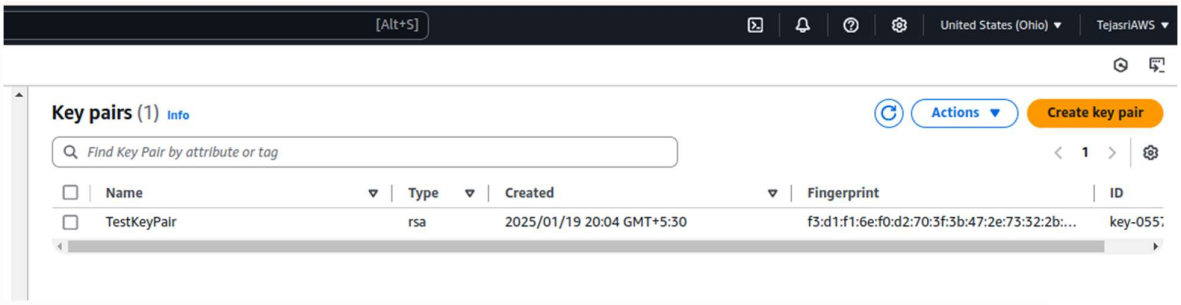
Verify that your account has been created successfully and you are familiar with the AWS Console (Note: don't do the conclusion part in the documentation provided).

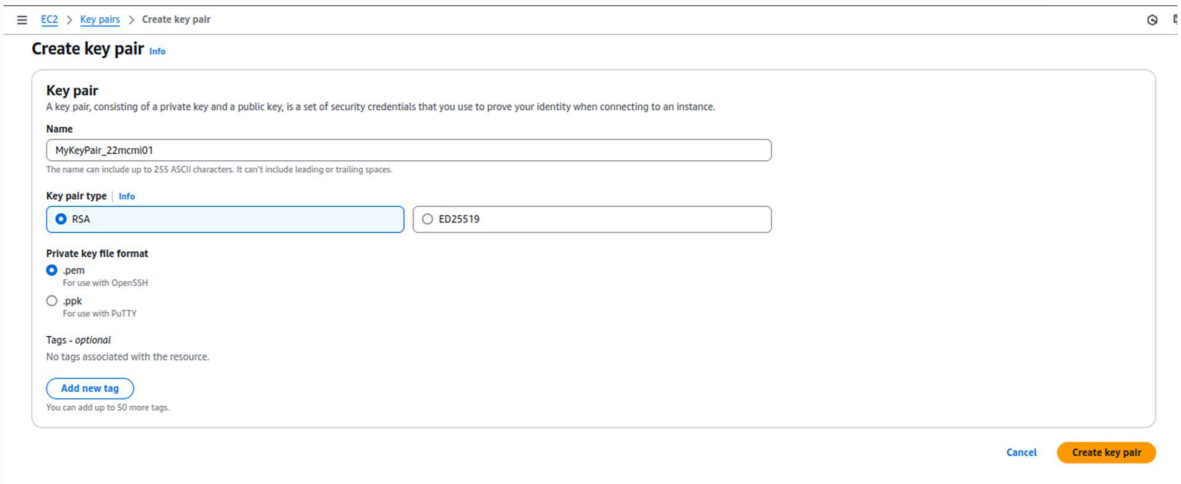
- Log into console.aws.amazon.com with your user credentials
- Create a ssh Key Pair for accessing Virtual Machines over the network. Make sure the name of the Key Pair includes your student id!

- Search for EC2 in the search bar, then at EC2 Dashboard, `Network and Security -> KeyPairs`



- Then create a new keypair



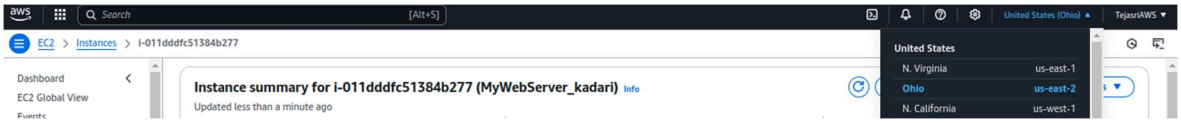


    - **NB**! This will download the private key as a text file into your computer with a `*.pem` extension. Copy the file into a location from where you can easily find it later.
- Take a screenshot of the key pairs page at the EC2 dashboard. Take a screenshot of the IAM dashboard, showing the security recommendations count to be zero. This shows you configured your MFA properly. **Note**: See that all your screenshots have your AWS account name displayed (This should identify you) (2 Marks)

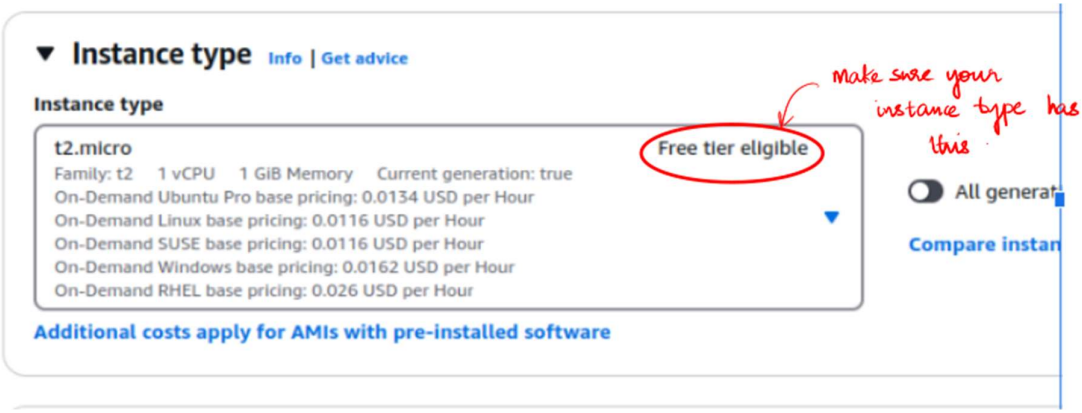
Exercise 2.2. - Requesting computing resources from the cloud

In this exercise you will start a Cloud instance (or virtual machine) while specifying its configuration and computing resources available for it.

- Use the AWS Cloud Console for this task
- Search for EC2 in the search bar, then at EC2 Dashboard, `Instances` -> `Launch Instances`
- **NB**: All of you use us-east-2 availability zone. If you keep changing these availability zones, you will be seeing the running instances from other availability zones. If you forget the instances running you may incur costs. So, please be careful here.



- Once you get into the “Launch Instances” page, you need to configure the following tasks.
    - Use your student id as the Instance Name under “Name and Tags” Label
    - Under “Application and OS Images (Amazon Machine Image) “ select Ubuntu Linux Image or “ any Free tier eligible Linux image”
    - Select the instance type to be “t2.micro”.
    - Note: “CAREFUL when choosing the instance type, choosing larger types incurs costs, stick to t2.micro which is **free tier eligible**”



    - In “KeyPair (Login)” tab select the keypair, as the one created in the Exercise 2.1
    - In the “Network Settings”, click edit
        - click the drop down “Auto-Assign Public IP”, keep it enabled, so that you can connect to it over the internet

        - In Firewall (security groups) -> click “create Security Group” -> Inbound security group -> Inbound security groups -> Type = “SSH” port is enabled by default (observe this) and the “Source type“ -> select “anywhere“
    - In the “Configure Storage”, you have a maximum of 30GB in the free tier list, so keep it to 8 GB.
    - Let the default values continue in the “Advanced details” section
    - Keep the number of instances to “1”
    - You can check the summary of the instance details before launching the instance.
    - Click “Launch Instance”

Exercise 2.3. Accessing your Cloud instance over the internet

We will use Secure Shell (ssh) protocol to log into the started instance over the internet. Instances in the cloud can have multiple IP addresses. Public IP can be used for accessing the instance from outside the cloud and Private IP can be used for accessing the instance from inside the cloud (from other instances).

NB: You cannot connect to the instances from the university network. So, please use your mobile networks to connect to your instances.

- In the Services → Compute →EC2 & its DashBoard → Instances, you can see the instances you have created in the previous section. On clicking the “instance id”, you would see the instance summary, where you can find the instance’s Public IPv4 DNS address/”Auto Assigned IP address”
- Log into the instance through ssh using SSH Key based authentication
    - On Linux:
        - `ssh -i path_to_my_key_pair_file ubuntu@<instance public ip address>`
        - For example: *ssh -i .cloud/srirama_key.pem ubuntu@172.17.64.63*
        - if you get an error, check that the path to the keyfile is correct and that it has correct rights (`chmod 400 <filename>`)
    - On Windows:
        - Either copy the private key pair file to the lab server (Note: You worked with it in Exercise 1) and use the previous ssh command.
        - Or use Putty, SSH secure shell, Git BASH command line, or WinSCP program to get a command line interface to a remote server through ssh.
            - When using Putty or SSH secure shell in windows we first have to transform the private key file (_keyname_.pem) we downloaded from AWS Cloud into a `.ppk` file.
                - PEM or Privacy Enhanced Mail is a Base64 encoded DER certificate. PEM certificates are frequently used for servers as they can easily be translated into readable data using a simple text editor.
        - If you are using `Putty` to connect to cloud instance with ssh you should use PuTTYgen to convert certificate into Putty specific `*.ppk` format. Use `Load` and `Save private key` functionality in the PuttyGen program to do it.
            - You can use the To prepare to connect to a Linux instance from Windows using PuTTY section @ https://www.earthdata.nasa.gov/learn/data-recipes/connect-your-ec2-instance-using-putty-v1-1 as a guide for converting the .pem key file into .ppk file.

        - Username for the SSH connection has to be `ubuntu`
            - Specify it in Putty under `Connection->Data->Login details`
        - Host must be the public IP of the instance you started
        - SSH Key must be the same `.ppk` key you converted with PyttyKeyGen
            - Specify its location under: `Connection->SSH->Auth->Private key file…`

NB: Take a screenshot of your home directory on the cloud instance. Also take a screenshot of the instance summary in the EC2 dashboard.

- On the home drive type the following command. `cat ~/.ssh/authorized_keys`
- Take a screenshot of it.
- The private IP in the instance description and the hostname displayed when you connect via ssh should match. The authorized keypair name and your student id should match. (2 Marks)

Exercise 2.4. Configuring Cloud Instance & installing software

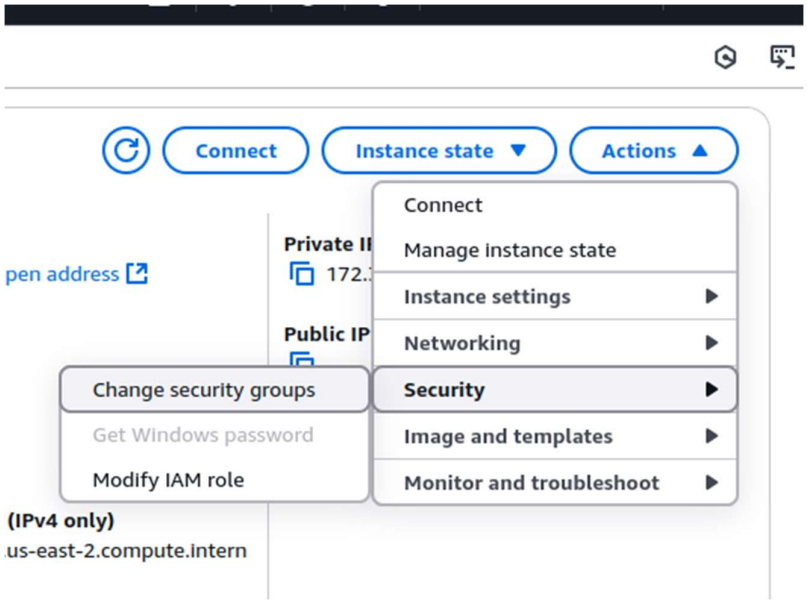
1. Install Apache web server on the instance
    - First we need to update the software package listings using apt-get or aptitude command:
        - `sudo apt update`
    - Install the apache2 package:
        - `sudo apt install apache2`
        - You might encounter an error stating that something is `locked` that is Ubuntu running some updates in the background. So please give it a few minutes to complete and try again later. (Use with caution! https://www.tecmint.com/fix-unable-to-lock-the-administration-directory-var-lib-dpkg-lock/)
2. Check that the installation of the Apache 2 web server is successful
    - Try accessing the public ip of your web server instance from a browser
        - This, in theory, should display the index.html web page that's being provided by your web server.
        - However, it does not work at the moment, as by default communication with the cloud instances from outside other than ssh (port 22) is restricted by default.
    - Let us use an alternative approach to verify our web server is running
        - Log into the instance through ssh
        - Use the `wget` command to download 'index.html' from your server.
        - On the instance command line run `wget localhost`.
            - This should download index.html file into the currently active directory, which among other html code should contain a string "It works!".
        - Use `less index.html` command to check the downloaded file content from the command line
        - You also tried curl. Use your knowledge from previous exercises.
        - Alternatively you can use command line web-browser `sudo apt install lynx` and `lynx localhost`
3. Modify (or replace) the current index.html file at `/var/www/html/index.html` to change the web page content.

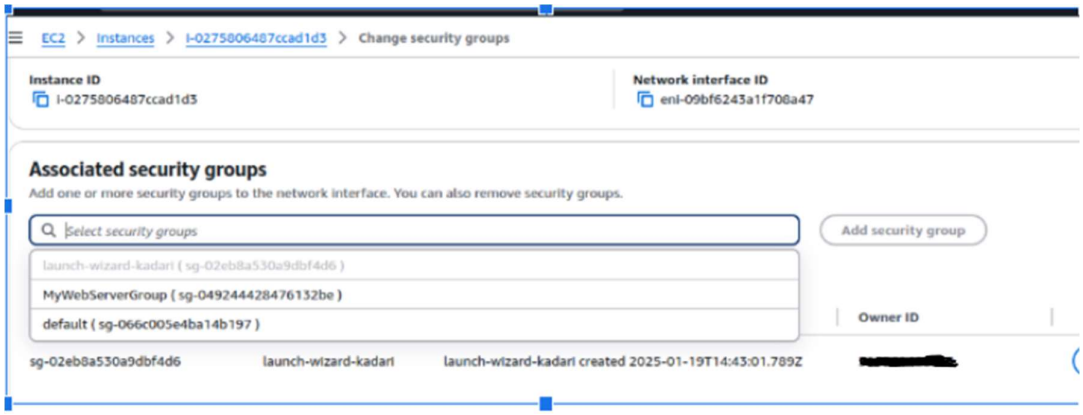
- How exactly you change its content is up to you, but it should at least contain your student id, so it is possible to visually see that you have modified it.
- Command line file editor `nano` can be used to modify file contents.
- You will also need to use `sudo` command as your user does not have permissions to modify this file otherwise.
- Feel free to replace the whole HTML file with a new one. (2 Marks)

Exercise 2.5. Creating Security group to enable access through port 80

To allow access to the hosted web server on your instance you need to create a new security group and define an access rule for the default HTTP port 80.

1. Under the EC2 DashBoard → Network Settings →Security Groups and create a new security group by clicking the `Create Security Group` button
2. Choose a `name for this security group`, which should include your student id.
   - Eg: WebServer_studentid
3. Add inbound rules → add rule → type[" select http"] & source ip (Anywhere-IPV4)
4. Create the security group.
5. Now to add this security group to your instance.
   - At the same EC2 Dashboard →Instances →click on the check box corresponding to the instance → actions → drop down select "Security" → change security groups.





- Select the security group created in step 2 of this exercise and then click “add security group”.

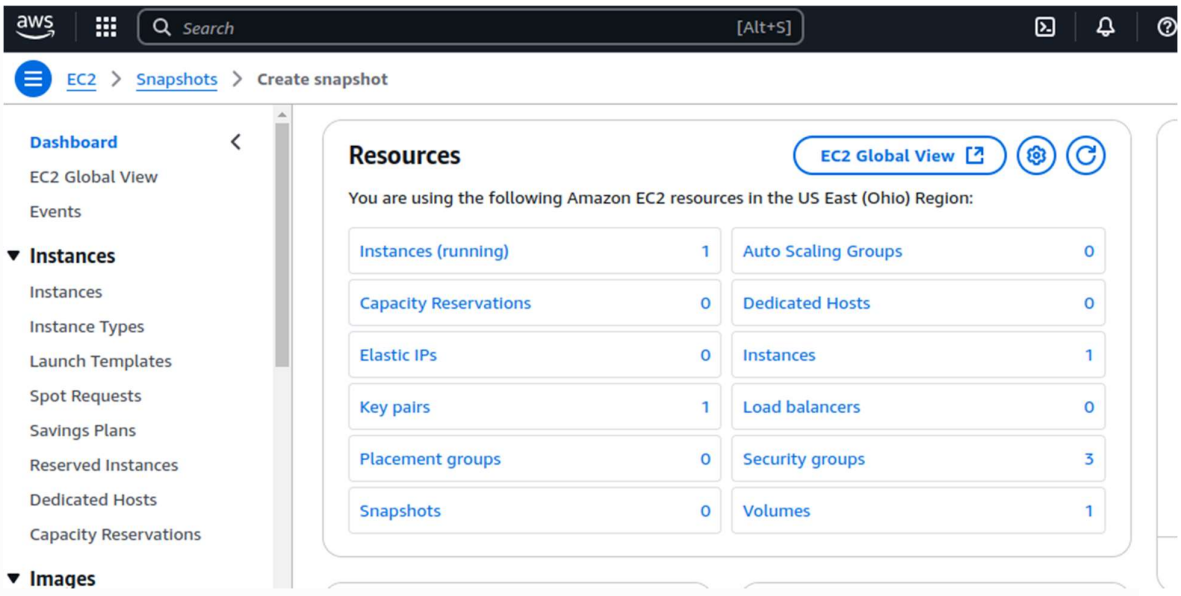
6. Access your instance through a web browser

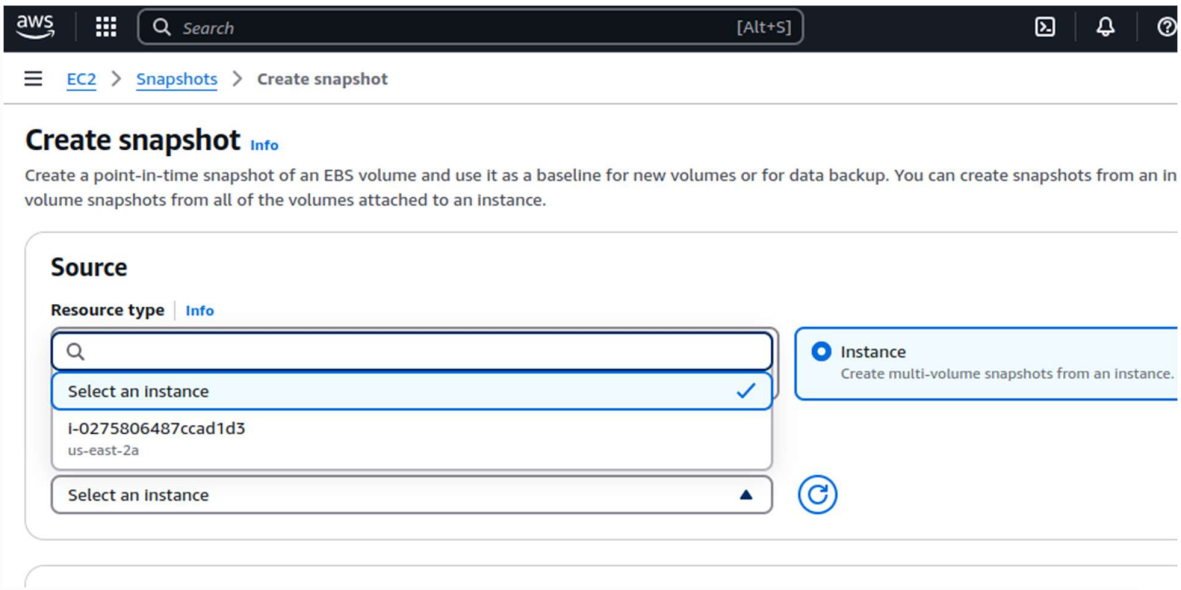
NB: Take a screenshot of your web server index file. (2 Marks)

Exercise 2.6. Creating a new Cloud Instance snapshot

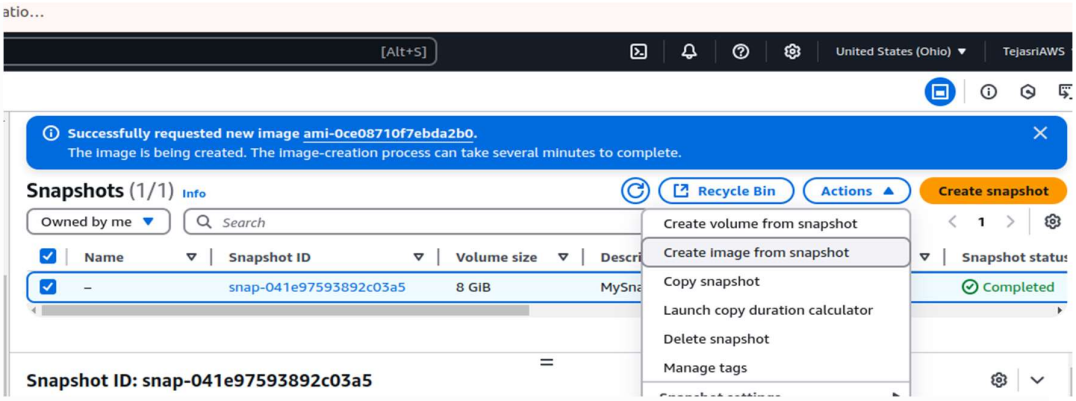
Lets save all the changes you have made to the instance by creating a snapshot. This allows you to start multiple already-configured Apache web server instances at once, with your web application already included.

- Under the EC2 Dashboard -> Snapshots -> create snapshot (or) EC2 Dashboard → Elastic Block Store →Snapshots.
- Here we are creating the snapshot of the instance. At the source, select the resource type to be “Instance”

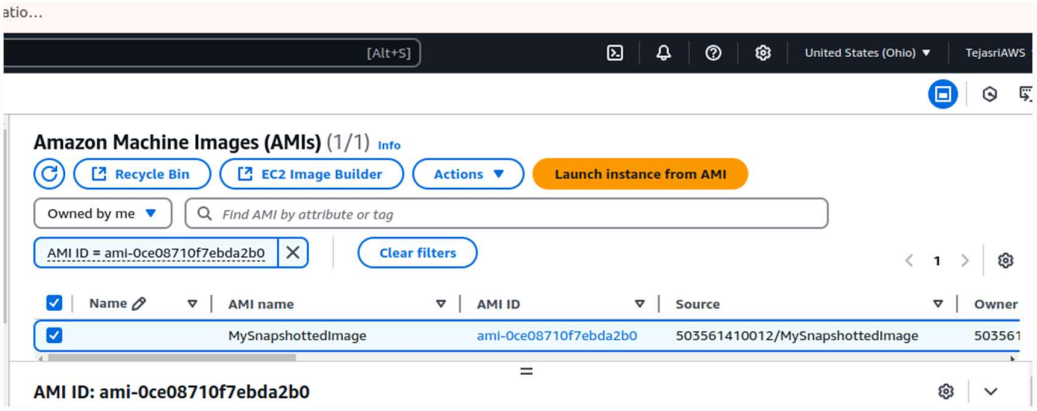




- Choose a name for this snapshot, it must include your studentid.
- After you confirm that the snapshot is ready, terminate your instance.
- Once the Snapshot is ready, create a image from the snapshot, and its name must include your studentid.



- Then you can verify the Amazon Machine Image created out of the snapshot, EC2 Dashboard → Images → AMI's



- Click on the AMI and create an instance out of the image. The instance creation process will be the same except that your “Application and OS Images” will use the AMI image created. Make sure you attach the security group you have already created for SSH and HTTP port access to this instance.
    - Make sure you can access the Apache index page from the web browser using your new instance's IP address.
    - Make a screenshot of the browser showing your deployed web page on the instance (2 Marks)
        - NB! Your name must be clearly visible on the screenshot you took in the previous task!
        - From the screenshot, the url with the public IP should be visible.
        - Also produce screenshots of your snapshot page and AMI page.

- In the future, you are able to start a copy (or multiple copies) of this web server at any time by starting a new instance from this snapshot.

You must delete the following after finishing your exercise

1. Your AWS Instance
2. Your Snapshot of the instance from the snapshots
3. Your amazon machine image that was created.

Follow these guidelines while submitting the exercises

1. Generate a PDF file with answers/screenshots with name being your studentid_EX2
2. Put a pdf file with the answers for the questions
    a. For programming tasks show screenshots of the program output for the correct and wrong inputs
    b. For commands tried, provide the screenshots
3. Do not copy code/answers from internet. If you copy, please leave a comment with the URL on the file
4. Plagiarism will not be tolerated and dealt through the formal channels of the university